

Demo of EITPose: Wearable and Practical Electrical Impedance Tomography for Continuous Hand Pose Estimation

Hongyu Mao*
Carnegie Mellon University
United States
hongyum@andrew.cmu.edu

Alexander Kyu*
Human Computer Interaction
Institute, Carnegie Mellon University
United States
akyu2@andrew.cmu.edu

Junyi Zhu
MIT CSAIL
United States
junyizhu@mit.edu

Mayank Goel
School of Computer Science, Carnegie
Mellon University
United States
mayankgoel@cmu.edu

Karan Ahuja
Computer Science, Northwestern
University
United States
kahuja@northwestern.edu

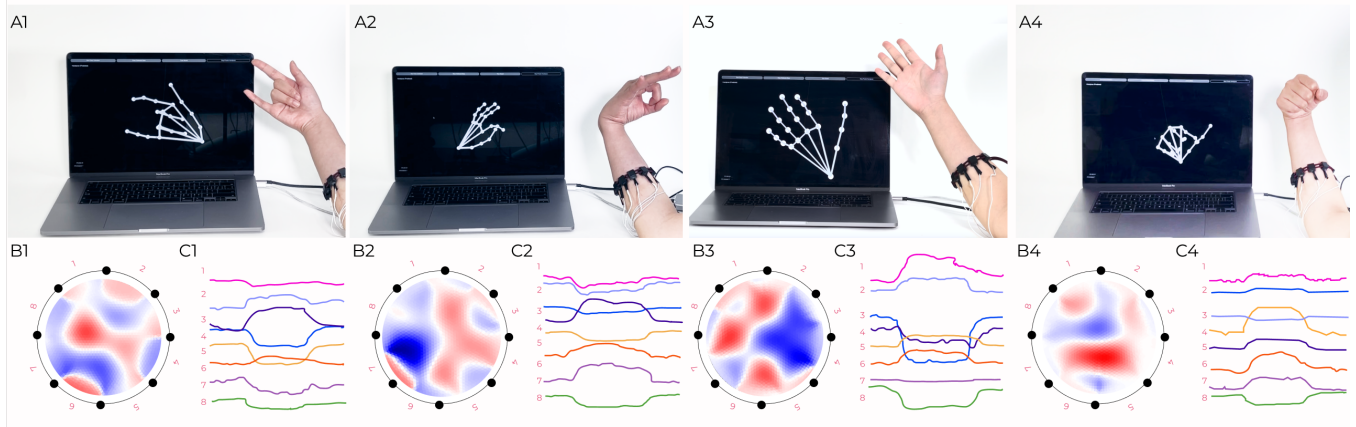


Figure 1: EITPose is a wrist-worn device for real-time hand pose estimation (A) that uses Electrical Impedance Tomography (EIT), to measure the impedance changes within the forearm. (B) shows a 2D reconstruction of this impedance map and (C) shows the average measured impedance signals per electrode emitter pair (1 through 8) while the user performs the hand pose.

Abstract

Real-time hand pose estimation has a wide range of applications spanning gaming, robotics, and human-computer interaction. In this paper, we introduce EITPose, a wrist-worn, continuous 3D hand pose estimation approach that uses eight electrodes positioned around the forearm to model its interior impedance distribution during pose articulation. Unlike wrist-worn systems relying on cameras, EITPose has a slim profile (12 mm thick sensing strap) and is power-efficient (consuming only 0.3 W of power), making it an excellent candidate for integration into consumer electronic devices. In a user study involving 22 participants, EITPose achieves with

a within-session mean per joint positional error of 11.06 mm. Its camera-free design prioritizes user privacy, yet it maintains cross-session and cross-user accuracy levels comparable to camera-based wrist-worn systems, thus making EITPose a promising technology for practical hand pose estimation.

ACM Reference Format:

Hongyu Mao, Alexander Kyu, Junyi Zhu, Mayank Goel, and Karan Ahuja. 2024. Demo of EITPose: Wearable and Practical Electrical Impedance Tomography for Continuous Hand Pose Estimation. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3672539.3686770>

*Both authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '24, October 13–16, 2024, Pittsburgh, PA, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0718-6/24/10

<https://doi.org/10.1145/3672539.3686770>

1 Introduction

Hand digitization has been extensively researched for decades, with applications in mixed reality, robotics, sports medicine, sign language, and spatial user interfaces. Most accurate hand-pose tracking methods use direct sensing, primarily camera-based solutions, which face issues like environmental noise[3], occlusion[1], and privacy concerns[5].

We introduce EITPose, a wrist-worn device using Electrical Impedance Tomography (EIT) to estimate hand poses continuously. EIT uses surface electrodes to measure impedance and generate cross-sectional images, useful for non-invasive and affordable medical imaging.

Previous research has explored wearable and mobile EIT form-factors for gesture classification, activity detection[6], and rehabilitation[4]. However, using a wrist-worn EIT band for accurate hand pose estimation presents challenges like poor electrode connectivity and varied body composition affecting measurements. EITPose addresses these with stable electrode connectivity and robust signal processing.

EITPose extends EIT-Kit[7], an open-source EIT toolkit, and includes features for optimal connectivity, signal-to-noise ratio, and a novel waveform checker algorithm for signal robustness. It also offers user-friendly enhancements like improved wireless capabilities and an adjustable band design for consistent electrode contact.

In this paper, we demonstrate the feasibility of continuous hand pose estimation using electrical impedance tomography. The improvements to sensor board, firmware, and band enables EITPose to achieve an accuracy comparative to camera-based methods while maintaining a low-power and privacy-preserving profile.

2 Implementation

2.1 Sensing Principle

Electrical Impedance Tomography (EIT) is an imaging technique that uses surface electrodes to inject alternating currents (AC) and measure voltages to assess impedance changes in an object. In our 8-electrode setup, we use a 4-pole configuration to measure up to 40 different impedance paths, providing detailed information on the internal structure and conductive properties of the forearm.

Our system includes an EIT sensing board and an elastic wristband made of eight stainless steel electrodes to ensure consistent contact across users.

2.2 Sensing Board Design

Building on the open-source EIT-kit, our design includes a stable current source for optimal injection current and measurement gain, enhanced by additional hardware and software filters. To mitigate variations in skin condition and physiological changes, we perform automated device initialization to optimize the injection current and voltage measurement gain for each electrode pair, ensuring connectivity and optimal signal-to-noise ratio. A waveform shape checker algorithm maintains signal quality by calculating the mean absolute error between measured data points and an ideal sine wave, using signals that pass the noise threshold for pose inference. An additional high-pass filter mitigates noise during MUX shifting, especially for frequencies below 3.68kHz, delivering cleaner signals and more reliable data.

2.3 Band Design

Our wristband, equipped with eight cylindrical stainless steel electrodes, ensures enhanced skin connectivity and adaptability to different forearm sizes. Mounted on 3D-printed PLA cubes connected to an elastic strap, the design allows for precise placement

and accurate impedance measurements. The electrodes interface with the customized EIT-kit sensing board through cold-welded wires.

Operating within a sample rate of 10 to 13 FPS, the device maintains stable signals at 10 FPS with the power consumption of 0.3W, significantly lower than previous methods[2][1], all while maintaining a compact design. In our user study with 22 participants, EITPose has a within-session mean per joint error of 11.06 mm.

3 Interaction and Demonstration

EITPose is highly interactive due to its compact form factor, low computational requirements, and lightweight design. To demonstrate EIT's capability, we have built a real-time live prediction system that processes EIT signals and visualizes predicted hand poses in real-time.

Real-Time Tomography Demonstration: Our system uses pyeit to convert raw Electrical Impedance Tomography (EIT) data into real-time reconstructive images, effectively displaying the internal tissue and bone distribution based on varying impedance values. This real-time tomography allows users to intuitively visualize the intricate internal structures of the hand as they move.

Demonstration of Real-Time Continuous 3D Hand Pose Tracking: For demonstrating continuous 3D hand pose tracking, we have developed a lightweight, real-time prediction system that processes raw EIT data using SciPy's ExtraTreesRegressor. The model outputs a vector of size 63, representing 21 3D hand pose key points, which are visualized in Python with updates at 10 Hz.

During the demonstration, users can intuitively learn how the EIT system detects varying impedance values to reconstruct images of the hand's internal structures. As users move their hands, they see their movements and poses translated into real-time 3D models. This setup provides an engaging, educational experience, demonstrating the principles of EIT sensing and the system's precise, responsive hand pose estimation, showcasing its potential for applications in virtual reality, robotics, and beyond.

4 Conclusion

We have presented EITPose, a mobile and self-contained form-factor arm band that enables continuous 3D hand pose estimation using Electrical Impedance Tomography (EIT) to capture the interior impedance geometry of a user's arm. Our approach allows for a minimally invasive (12 mm) and power-efficient (0.3 W) form factor, showcasing the potential for integration into consumer electronic form factors. Its camera-less approach enables a privacy-sensitive sensing system but with the cross-session and cross-user accuracy profiles of camera-based wrist worn systems, thus providing the best of both worlds.

References

- [1] Nathan Devrio and Chris Harrison. 2022. DiscoBand: Multiview Depth-Sensing Smartwatch Strap for Hand, Body and Environment Tracking. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 56, 13 pages. <https://doi.org/10.1145/3526113.3545634>
- [2] Fang Hu, Peng He, Songlin Xu, Yin Li, and Cheng Zhang. 2020. FingerTrak: Continuous 3D Hand Pose Tracking by Deep Learning Hand Silhouettes Captured by Miniature Thermal Cameras on Wrist. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 4, 2, Article 71 (jun 2020), 24 pages. <https://doi.org/10.1145/3397306>

- [3] Daehwa Kim and Chris Harrison. 2022. EtherPose: Continuous Hand Pose Tracking with Wrist-Worn Antenna Impedance Characteristic Sensing. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 58, 12 pages. <https://doi.org/10.1145/3526113.3545665>
- [4] Evan A. Susanto, Raymond KY Tong, Corinna Ockenfeld, and Newmen SK Ho. 2015. Efficacy of robot-assisted fingers training in chronic stroke survivors: a pilot randomized-controlled trial. *Journal of NeuroEngineering and Rehabilitation* 12, 42 (2015). <https://doi.org/10.1186/s12984-015-0033-5> arXiv:<https://doi.org/10.1186/s12984-015-0033-5>
- [5] Hui-Shyong Yeo, Erwin Wu, Juyoung Lee, Aaron Quigley, and Hideki Koike. 2019. Opisthenar: Hand Poses and Finger Tapping Recognition by Observing Back of Hand Using Embedded Wrist Camera. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 963–971. <https://doi.org/10.1145/3332165.3347867>
- [6] Yang Zhang and Chris Harrison. 2015. Tomo: Wearable, Low-Cost Electrical Impedance Tomography for Hand Gesture Recognition. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology* (Charlotte, NC, USA) (UIST '15). Association for Computing Machinery, New York, NY, USA, 167–173. <https://doi.org/10.1145/2807442.2807480>
- [7] Junyi Zhu, Jackson C Snowden, Joshua Verdejo, Emily Chen, Paul Zhang, Hamid Ghaednia, Joseph H Schwab, and Stefanie Mueller. 2021. EIT-Kit: An Electrical Impedance Tomography Toolkit for Health and Motion Sensing. In *The 34th Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '21). Association for Computing Machinery, New York, NY, USA, 400–413. <https://doi.org/10.1145/3472749.3474758>